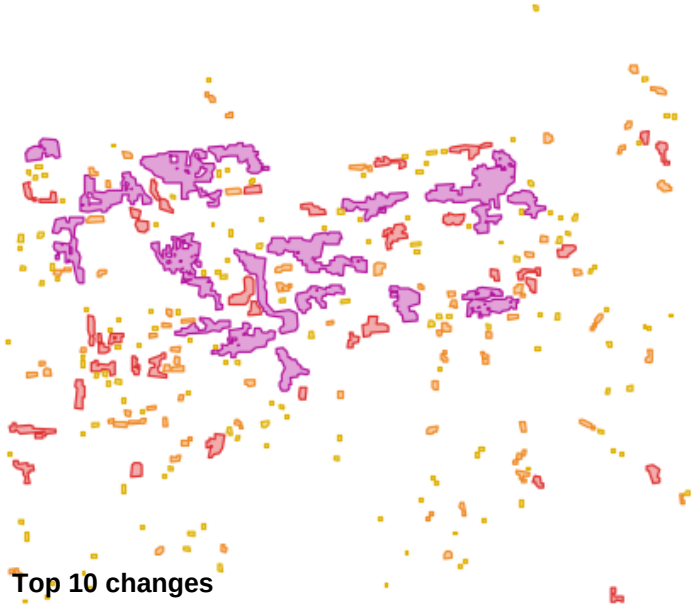
GEOAI 04 change report: Navi Mumbai airport build-out



Totals

Raw detections: 2185
Real changes after filters: 265
Area changed: 250.69 ha
Cropland affected: 156.89 ha (upper bound)
Built-up land affected: 2.51 ha
critical: 17
high: 35
medium: 69
low: 144

Open this result: /scene/navi-mumbai-airport

Top 10 changes

#	Tier	Type	Area (ha)	Conf.	Reason
1	critical	built_up_gain	21.585	0.904	new construction / built-up: SWIR/blue ratio (paved surfaces are flatter) fell (
2	critical	built_up_gain	16.955	0.94	new construction / built-up: built-up index fell (-0.05), water index rose (+0.0
3	critical	built_up_gain	15.855	0.892	new construction / built-up: SWIR/blue ratio (paved surfaces are flatter) fell (
4	critical	built_up_gain	13.53	0.934	new construction / built-up: SWIR/blue ratio (paved surfaces are flatter) fell (
5	critical	water_loss	9.61	0.966	water receded: built-up index rose (+0.17), vegetation index rose (+0.22), water
6	critical	built_up_gain	10.79	0.933	new construction / built-up: water index rose (+0.05), SWIR/blue ratio (paved su
7	critical	built_up_gain	7.995	0.936	new construction / built-up: vegetation index fell (-0.11), water index rose (+0
8	critical	built_up_gain	8.25	0.923	new construction / built-up: water index rose (+0.06), SWIR/blue ratio (paved su
9	critical	built_up_gain	7.235	0.909	new construction / built-up: SWIR/blue ratio (paved surfaces are flatter) fell (
10	critical	built_up_gain	6.595	0.91	new construction / built-up: water index rose (+0.07), SWIR/blue ratio (paved su

Sources and limits

- Copernicus Sentinel-2 L2A (ESA), via the Element84 Earth Search COG bucket
- ESA WorldCover 2021 v200 (10 m), CC-BY 4.0
- Optical imagery cannot see through cloud; flagged areas with cloud are removed, not analysed.
- Change type comes from spectral index rules (no training); check critical changes on the ground.
- 10 m pixels cannot resolve single small buildings or people.
- Land cover is ESA WorldCover 2021, newer than the before image (2019): land already cleared or under construction can be mislabelled as cropland, so treat cropland as an upper bound
- Buildings and road segments need OpenStreetMap, which was not reachable when this result was built: shown as unknown, not zero